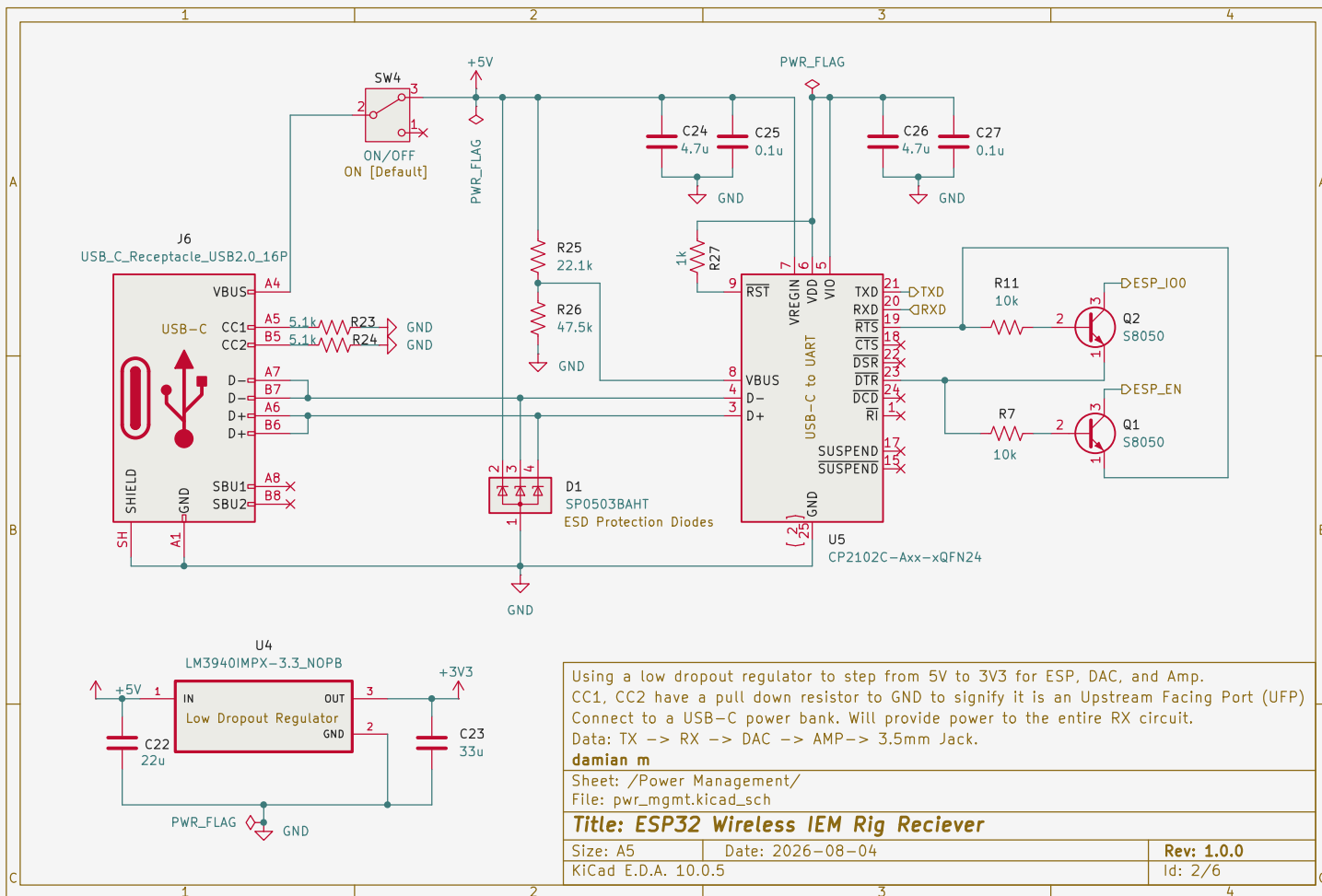






Using a low dropout regulator to step from 5V to 3V3 for ESP, DAC, and Amp. CC1, CC2 have a pull down resistor to GND to signify it is an Upstream Facing Port (UFP) Connect to a USB-C power bank. Will provide power to the entire RX circuit. Data: TX -> RX -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Power Management/

File: pwr_mgmt.kicad_sch

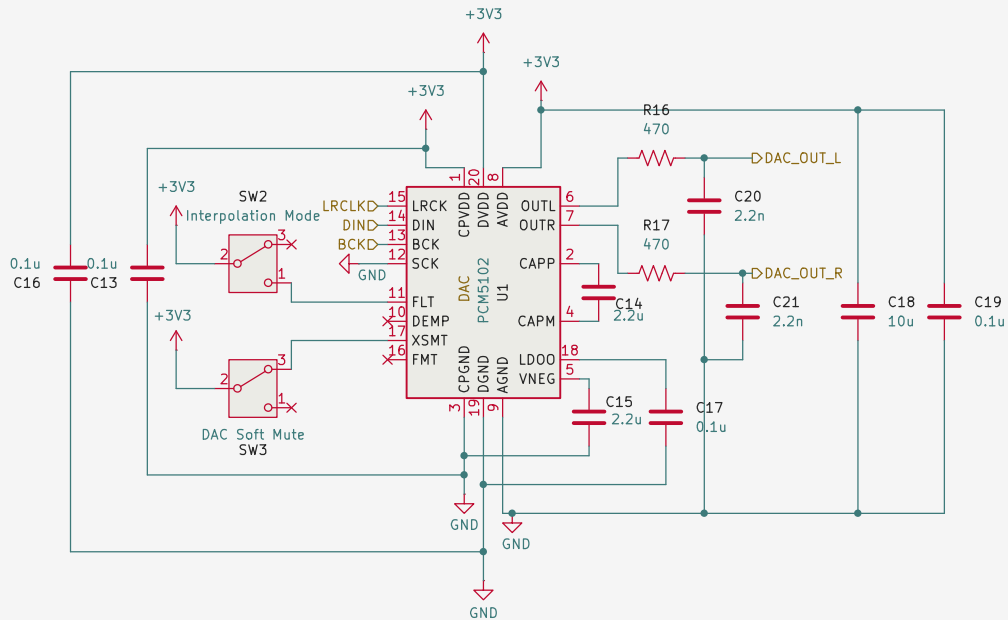
Title: ESP32 Wireless IEM Rig Reciever

Size: A5 Date: 2026-08-04

KiCad E.D.A. 10.0.5

Rev: 1.0.0

Id: 2/6



Use PLL functional diagram since we are grounding SCK

DEMP Low = DISABLE De-emphasis eq (44.1kHz).
High = ENABLE De-emphasis eq (44.1kHz).

FMT Low = I2S Format.
High = Left justified.

Soft Mute: XSMT LOW = Muted. XSMT HIGH = Unmuted. [Default]
Interpolation mode: FLT HIGH = Low Latency. FLT LOW = Normal Latency [Default]
Data: TX -> RX -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /DAC/

File: dac.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

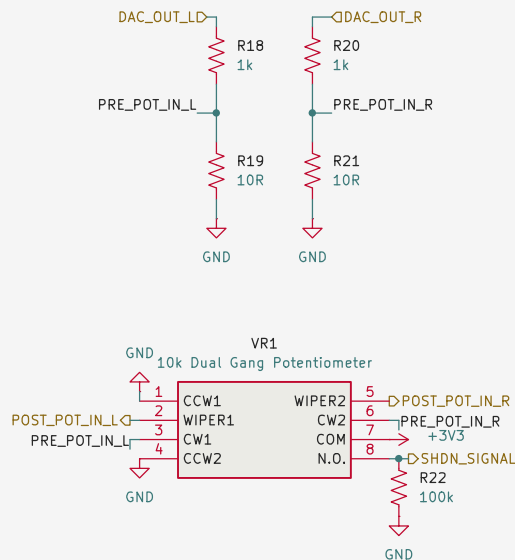
Size: A5

Date: 2026-08-04

Rev: 1.0.0

KiCad E.D.A. 10.0.5

Id: 4/6



POST_POT_IN_R/L becomes the input to the amp.
 Volume Knob with integrated off switch to kill amp from playing any audio.
 AMP_IN_L/R follows a voltage divider to lower volume to safer levels before hitting AMP.
 Data: TX -> RX -> DAC -> AMP-> 3.5mm Jack.

damian m

Sheet: /Volume Control/

File: vol_ctrl.kicad_sch

Title: ESP32 Wireless IEM Rig Reciever

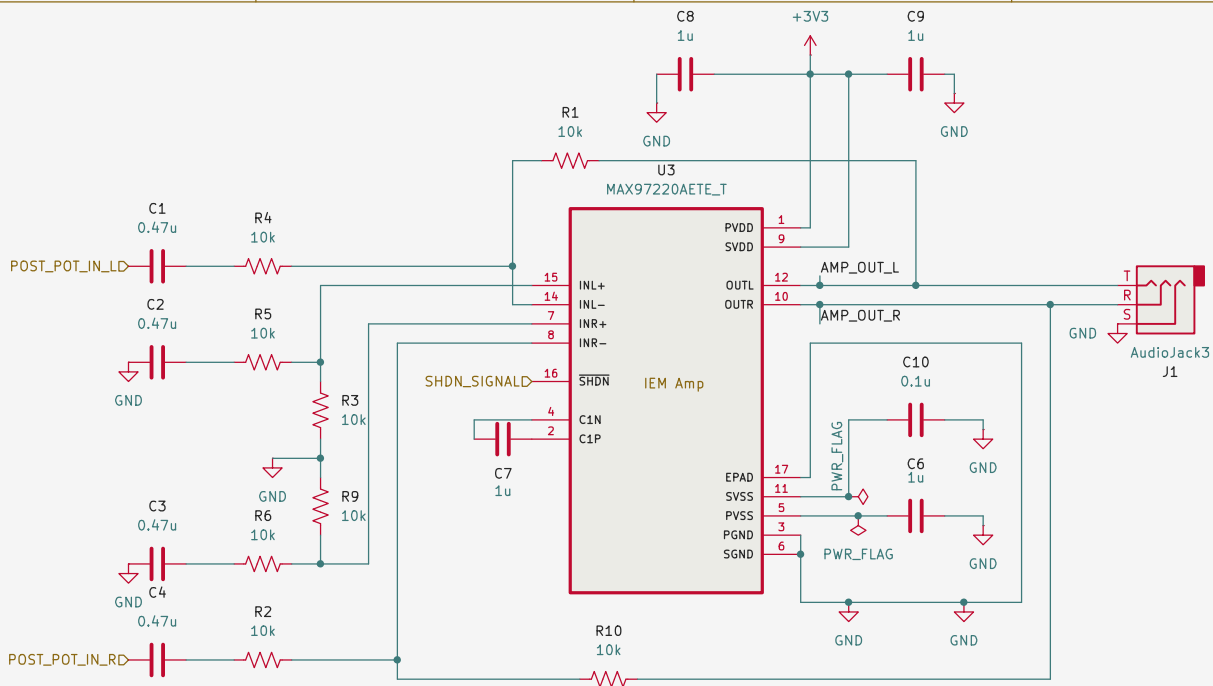
Size: A5

Date: 2026-08-04

Rev: 1.0.0

KiCad E.D.A. 10.0.5

Id: 5/6



SHDN_SIGNAL LOW acts as setting the volume to a "true 0%" by shutting off the IEM amp.

This is in contrast to the DAC Soft mute which acts as a mute toggle without changing the volume.

Amp follows the pseudo-differential input configuration because the DAC output is a single-ended input.

SHDN_SIGNAL is handled by the dual gang pot. N.O. = AMP OFF.
BIAS is in the same position as SVSS. Refer to in manual as BIAS.
Data: TX -> RX -> DAC -> AMP -> 3.5mm Jack.

damian m

Sheet: /Amp/
File: amp.kicad_sch

Title: ESP32 Wireless IEM Rig Receiver

Size: A5 Date: 2026-08-04
KiCad E.D.A. 10.0.5

Rev: 1.0.0
Id: 6/6